



BEYOND
AND
FINAL
TO
ROAD



<https://uobst.ilm.gg>

UOB
SUSTAINABLE
TRANSPORT

OUR TEAM

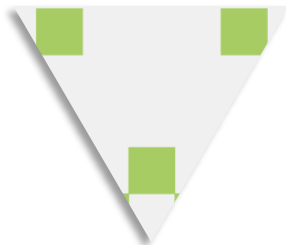
Meet the team members and maybe
remember their names

TEAM MEMBERS



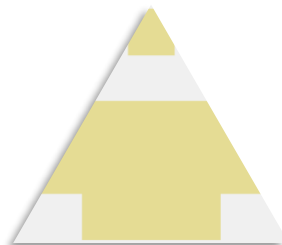
YIDI AI

Backend Specialist



ERIK MALTBY

Client Liaison



ALAN YANG

New Lens



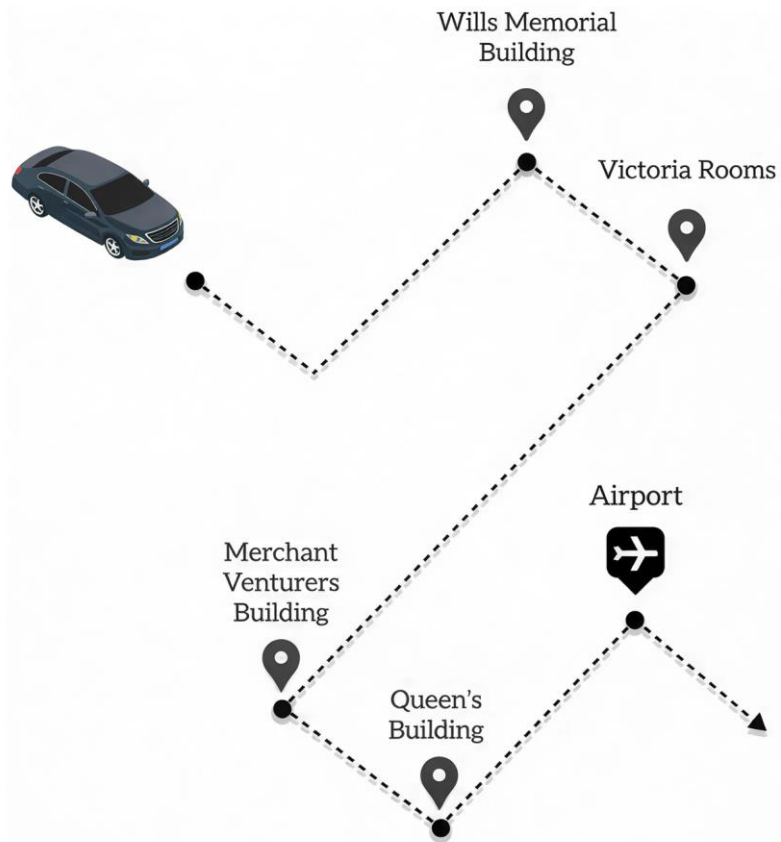
IOAN MOIR

Project Manager



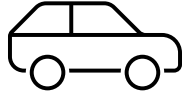
OG

Repo Polisher



01 PROJECT BRIEF

What are we working on ?



OUR CLIENT



University of
BRISTOL



PROLINE

A company that offers **professional, reliable,** and **affordable luxury taxi & chauffeur** services.



COLLABORATION

Proline has been offering **taxi services** to the University of Bristol **staff** since **2019**.



PROBLEM

The **current booking process** with the University of Bristol is **inefficient, time-consuming,** and **heavily manual**.



SOLUTION

Streamlining the manual process with a **digital experience,** promoting clear **communication** and **automation**.



PASSENGER



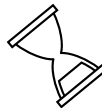
I want the booking approval process to be faster and transparent to **save time**.



FINANCE STAFF



I want to view the key details of a booking instantly and approve or reject it **quickly**.



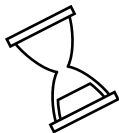
USER STORIES



PROLINE



We want to automate the current laborious back-and-forth booking process to **save time**.





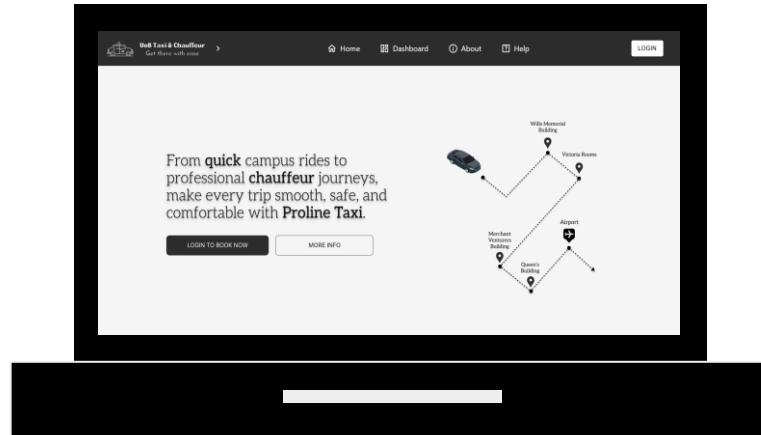
Visit the website
yourself and have
look around !

DEMO

Test Login Details

demo@example.com

demo1



Let's see what it looks like!

 <https://uobst.ilm.gg>

03

PROJECT ARCHITECTURE

What tools did we use ?



React

+



BACKBONE.JS

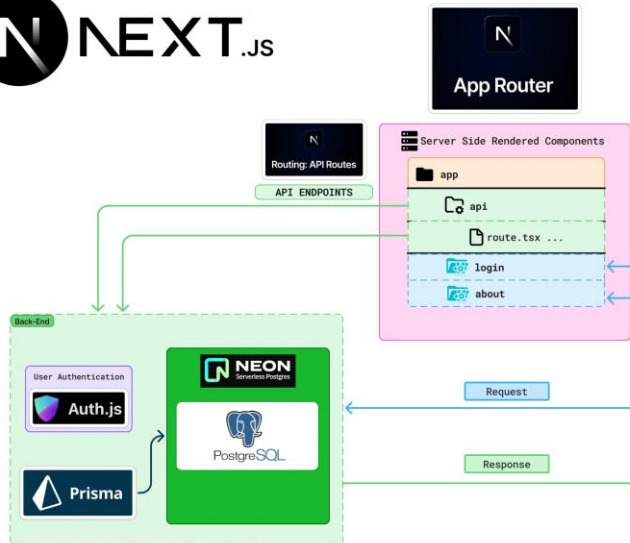
+



Semantic U



ANGULARJS



N NEXT.JS

- File based routing system.
- Server-Side Rendering (SSR).
- API Endpoint routes (Backend).

🐘 POSTGRESQL X NEON

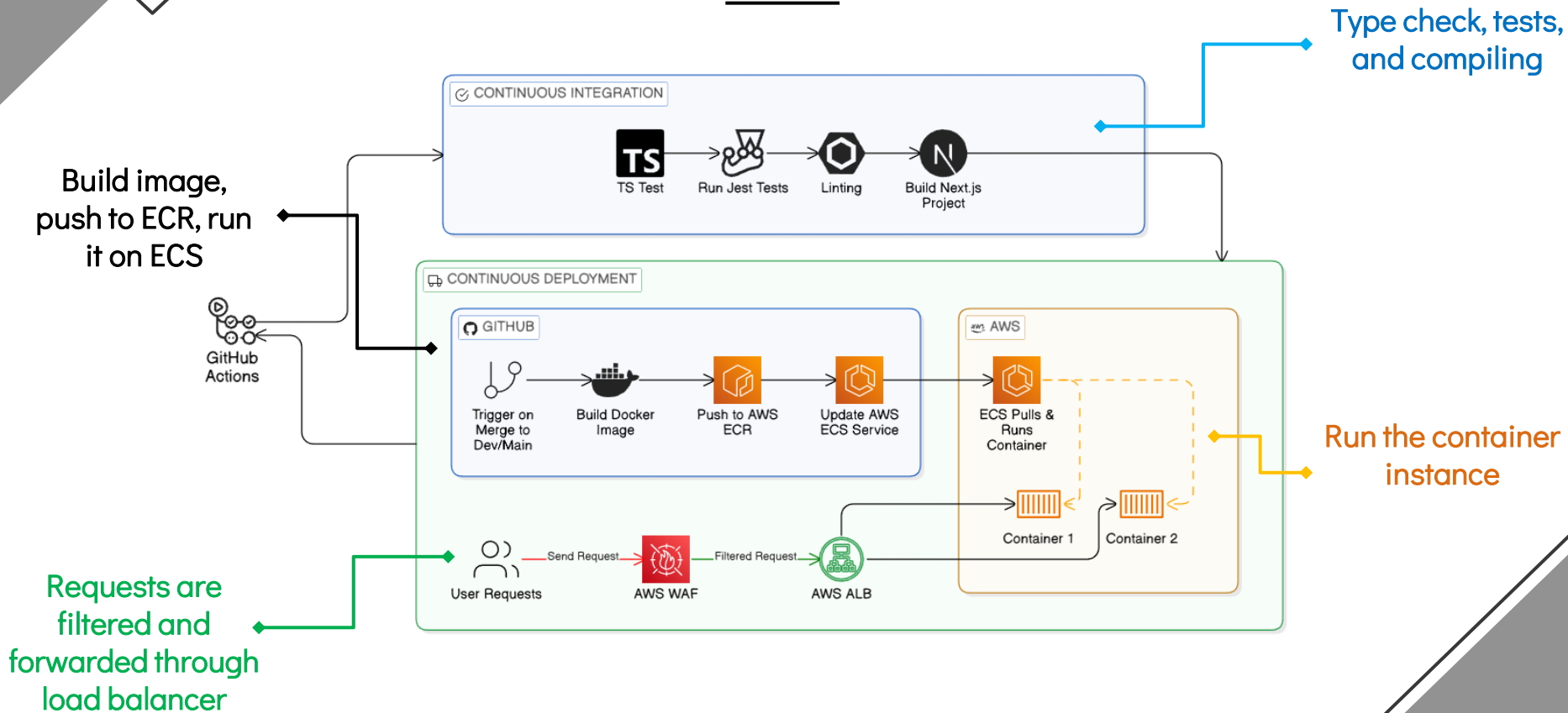
- Better concurrency
- Convenient GUI (Neon Console).
- Git-like branches for DB version control.



PRISMA

- JS-like query syntax.
- GraphQL adjacent data fetching.
- Layer of abstraction.

CLOUD ARCHITECTURE (CI/CD)





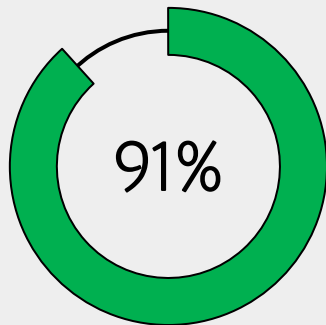
TEST COVERAGE



Frontend: 308 tests

Backend: 209 tests

Total: 517 tests



FRONT-END
COVERAGE

File	% Stmts	% Branch	% Lines
All files	91.46	83.38	91.46

The result is generated by Jest's built-in
coverage analyzer



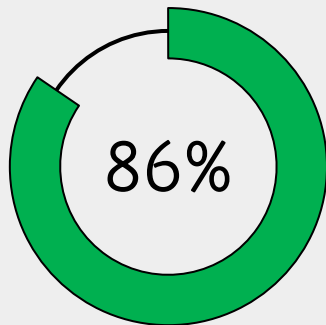
TEST COVERAGE



Frontend: 308 tests

Backend: 209 tests

Total: 517 tests



BACK-END
COVERAGE

File	% Stmts	% Branch	% Funcs	% Lines
All files	96.29	86.23	98.75	96.29



Branch logic coverage was chosen as the primary metric, as it best reflects backend behavior.

The result is generated by Jest's built-in coverage analyzer



04

TESTING DAY

Feedback and Critique
time



SURVEY QUANTITATIVE DATA

DEPLOYED SURVEY FRAMEWORK: [PSSUQ](#)

 27 RESPONSES 1 – GOOD 7 – BAD

1.83



SYSTEM USEFULNESS
(ease of use, intuitiveness)

2.11



INFORMATION QUALITY
(documentation, clarity of feedback)

1.68



INTERFACE QUALITY
(design, color scheme, elements)

OVERALL: 1.93 / 7

BOOKING DETAILS

Trip details:

Common pick-up locations

Please pick one.

Manually Enter ☐ Flight ☐ Via ☐

Drop-off location

Temple Quarter Enterprise Campus, Bristol

Please enter a drop-off location.

Pick-up date and time

20/03/2026

Please select a Time.

☒ Return trip

Return Trip Pick-up location

Return Trip Drop-off location

Enter

Return trip pick-up date and time

19/03/2026

Please select a return Time.

Lead passenger details:

I am the lead passenger ☐

Passengers

1

Additional information

Enter any additional information...

CONFIRM BOOKING

LOG IN

Email

soggywaffles@gmail.com

Password

Incorrect mail or password. Please enter the correct details or wait for your account approval.

Forgot Password?

LOG IN

Don't have an account? [Sign up](#)



PROLINE



University of BRISTOL

Proline Taxi | University of Bristol

This service, in partnership with Proline Taxi, aims to streamline the taxi booking process by automating the processes from requesting, approving, and even merging taxi services together. The platform also provides automated invoicing to simplify finances and clarify billing for your departments.

BOOKING MADE EASY

1. Fill in the booking form

2. Get approval

3. Receive confirmation and details via SMS

4. Off you go!

Check out proline taxi Reviews

John Doe
Great experience.

Allen Key
Very smooth travel.

Jane Doe
Love it!



PROLINE



University of BRISTOL

Proline Taxi | University of Bristol

This service, in partnership with Proline Taxi, aims to streamline the taxi booking process by automating the processes from requesting, approving, and even merging taxi services together. The platform also provides automated invoicing to simplify finances and clarify billing for your departments.

BOOKING MADE EASY

Simple steps to your ride

1. Fill in the booking form
Quick and easy form with all necessary details.

2. Get approval
Fast automated approval from your department

3. Receive confirmation
Get all details and driver info via SMS

4. Off you go!
Enjoy your comfortable, reliable ride

Check out Proline Taxi Reviews

Vanessa Rolland
Always helpful and on time, as well as very comfortable cars!

Thomas Lanzoni
Very professional, sympathetic and flexible!! recommend it!

Helen Elliott
Always an excellent service and great at keeping in touch with taxi on route.

Need Help?

+44 790 445 9504

sales@prolinetaxi.com

OPEN USER FEEDBACK

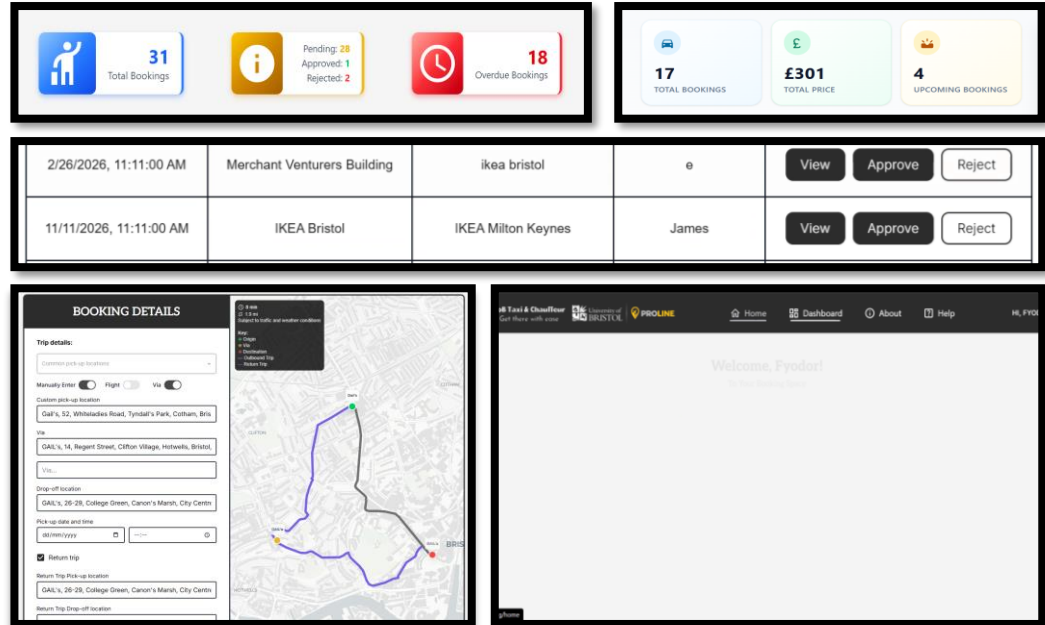


»»» ONLOAD MOTION ANIMATION

📍 CLEAN MAP UI WITH ROUTING

🖱️ BUTTONS WITH HOVER FEEDBACK

📄 NEAT INFO CARDS





SUGGESTIONS



MISLEADING

A lot of element have **unnecessary hover effects**, feel like **buttons**.



CONVENIENCE

Updating the map on **enter** rather than requiring clicks, interactive **card filtering**.



SECURITY

Profile **editing** should **prompt a password** entry.



UI/UX

Consistently placed **navbar** and **refreshing errors** after wrong input.

SUGGESTIONS IMPLEMENTED

MISLEADING

A lot of element have unnecessary hover effects, feel like buttons.



CONVENIENCE

Hitting enter should update the map, interactive card filtering.



SECURITY

Profile editing should prompt a password entry.



UI/UX

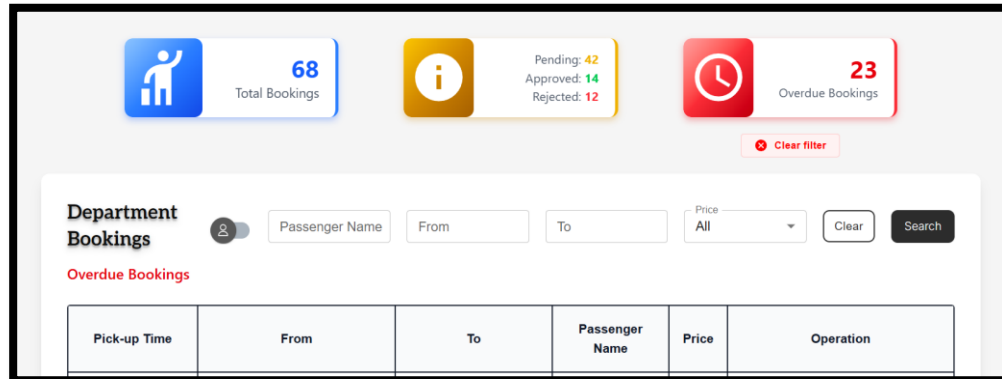
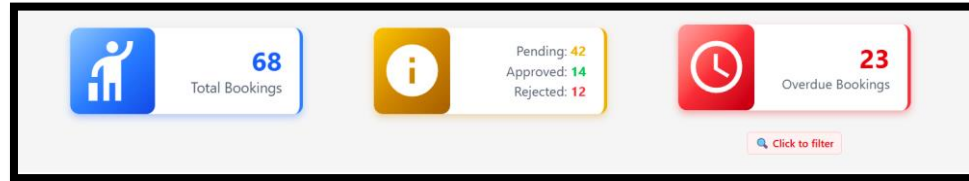
Sticky navbar and refreshing errors after wrong input.



SUGGESTIONS

CONVENIENCE

Hitting enter should update the map,  interactive card filtering.



The interface includes a header with three cards (Total Bookings: 68, Status Summary, Overdue Bookings: 23) and a "Clear filter" button. Below is a "Department Bookings" section with a search bar and filters for Passenger Name, From, To, and Price. A table titled "Overdue Bookings" is shown at the bottom.

Pick-up Time	From	To	Passenger Name	Price	Operation
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SUGGESTIONS

SECURITY

Profile **editing** should
prompt a **password** entry.



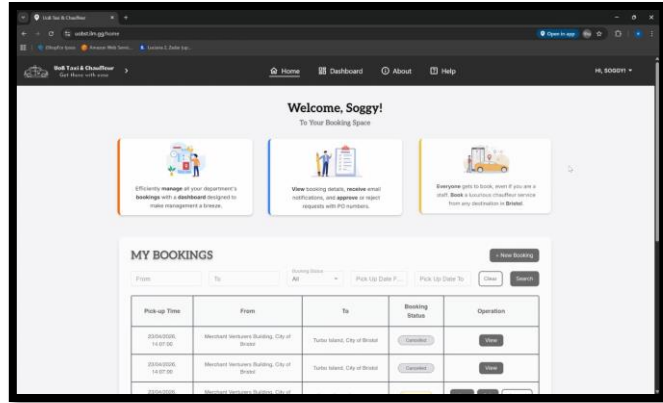
The screenshot shows a user profile editing interface. The background form is dimmed and contains the following fields:

- FULL NAME: Fyodor Tolstoi
- Contact Information:
 - EMAIL: fyodor@gm
 - Phone Number (with a red outline)
- Account Det (partially visible)
- DEPARTMENT: Economics

At the bottom of the form are two buttons: "Save Changes" and "Cancel".

Overlaid on top of the form is a white modal dialog box with the title "Enter Password to Save Changes". Inside the dialog is a password input field labeled "Password" with an eye icon for toggling visibility. Below the input field are two buttons: "Cancel" and "Save".

TASK BASED USABILITY TESTING



QUALITATIVE

- "I'm very happy with the profile side of the system"
- "It is very intuitive and looks very professional"
- "No suggestions for address input?"
- "Line by line address input would be better"

QUANTITATIVE

- 75% of tasks were completed **independently**, with the overall success rate being **100%** when **assistance** was given.
- All tasks were completed within **1 minute and 28 seconds**.

Time per Task and Task Success rate

Participant	Task 1	Task 2	Task 3	Task 4	Task 5
Participant 1	00:14	03:14	00:24	03:33	00:22
Participant 2	00:48	04:45	00:42	03:07	00:37
Participant 3	00:17	01:41	00:28	02:49	00:11
Participant 4	00:32	02:16	00:20	02:10	00:30
Mean Time	00:28	02:59	00:29	02:55	00:25

TASK BASED USABILITY TESTING

Recommendations for continued development

The following are recommendations given the analysis of user feedback, the recommendations given are not conclusive and further considerations would need to be made as to how to implement them into the project.

All users experienced no friction when logging in or switching accounts, additionally users didn't face any issues when completing the miscellaneous tasks given to them. Therefore, there will be no recommendations as to these tasks directly.

Many users experienced some friction with the booking page; this friction could be lowered by:

- Implementing auto complete on the address fields.
- Separating the address field to multiple sections (like online booking platforms).
- Removing the common pickup location dropdown or allowing for a user to type directly into it as if it were a text field, therefore removing the need for the "manually enter" toggle.
- Improve failed form submission errors, especially the false highlighting of the final field in the form on unrelated errors.
- When the "I am the lead passenger" toggle is selected, the passenger details should remain visible however be greyed out preventing a user from booking a taxi under the wrong name.
- Before submission, give a booking summary to the user allowing them to catch unwanted errors.
- The labels at the top of the home page for normal users should be interactable or present in a way as to not confuse the user into thinking they are buttons.

Lots of users experienced friction when completing the Booking approval/rejection task, some recommendations would be:

- The landing page of the finance staff accounts should be the dashboard page to prevent unnecessary confusion and to prioritise finance staffs' most likely action.
- Strong consideration should be taken as to whether finance staff accounts need the ability to book taxis. Removing this feature would remove confusion as to the differences between the home and dashboard page.
- To extend of the last recommendation a "switch account" or multiple account management system could allow finance staff to switch accounts to book taxis faster.
- The search filters could dynamically apply allowing users to identify mistakes in their search queries as they make them.
- Refinement of the date "clear field" button to be less visually striking.
- Finance staff should be given a text field upon rejecting a booking allowing the system to relay the reason for booking rejection, reducing manual communication.

- All tasks were completed within 1 minute and 28 seconds.

Review of the study

The study was a broad success, highlighting pitfalls in usability, however some key alterations could be made to improve the relevance and quality of the findings.

The study utilised proxy users, whilst the users who participated were all university staff, none of them were responsible for taxi booking administration. This could have a noticeable effect on the outcome of the study as they would have a better understanding of the differentiation between the booking creation and administration workflows.

Similarly, due to time constraints only four participants were able to complete the study. Whilst valuable feedback was collected in the qualitative findings the quantitative conclusions lacked academic relevance. Increasing the sample size would allow further statistical analysis of the time and success rate of the user group.

Alongside more participants, the study could be improved by a wider range of tasks covering all workflows within the application. To extend this idea, randomised scenario generation could be implemented using Large Language Models to eliminate the need for manual task creation for a large study sample size.

Whilst all staff were given the option to use a mouse, an improvement to the method of the study could be if participants could use their work computer when participating. This would allow the participant to be familiar with their peripherals, resulting in an increase in real-world accuracy in time per task measurements.

In an ideal scenario, it would be more appropriate to compensate participants for their time and feedback. This would have increased the participants commitment to the project and study. Unfortunately, this wasn't an option as the project is unfunded.

Throughout conducting the study, I made subtle but substantial changes to how I led the sessions, I became clearer when I explained the tasks vocally and gave better clarifications to tasks without unintentionally providing hints to the users.

As well as the form error mentioned earlier, when logging-in with the normal user account details provided, a google chrome alert triggered regarding the password being involved in a data breach. This distracted users from the tasks and interrupted the user experience and therefore test accounts should have passwords which aren't going to cause this to happen.

Reviewing all the videos took a lot of time, the review methods should be improved if the study were to be repeated with a larger sample size. Research should be conducted with the assistance of tools which would aid this process. The sessions should be adapted for these tools if

Study method

The study was designed for the primary users of our platform: university department finance staff. However, despite extensive communication, I was unable to recruit any finance staff to complete the study. Consequently, I recruited other university staff with experience in similar systems; unfortunately, these participants were unfamiliar with the current booking procedure agreed upon by Proline Taxi and the University.

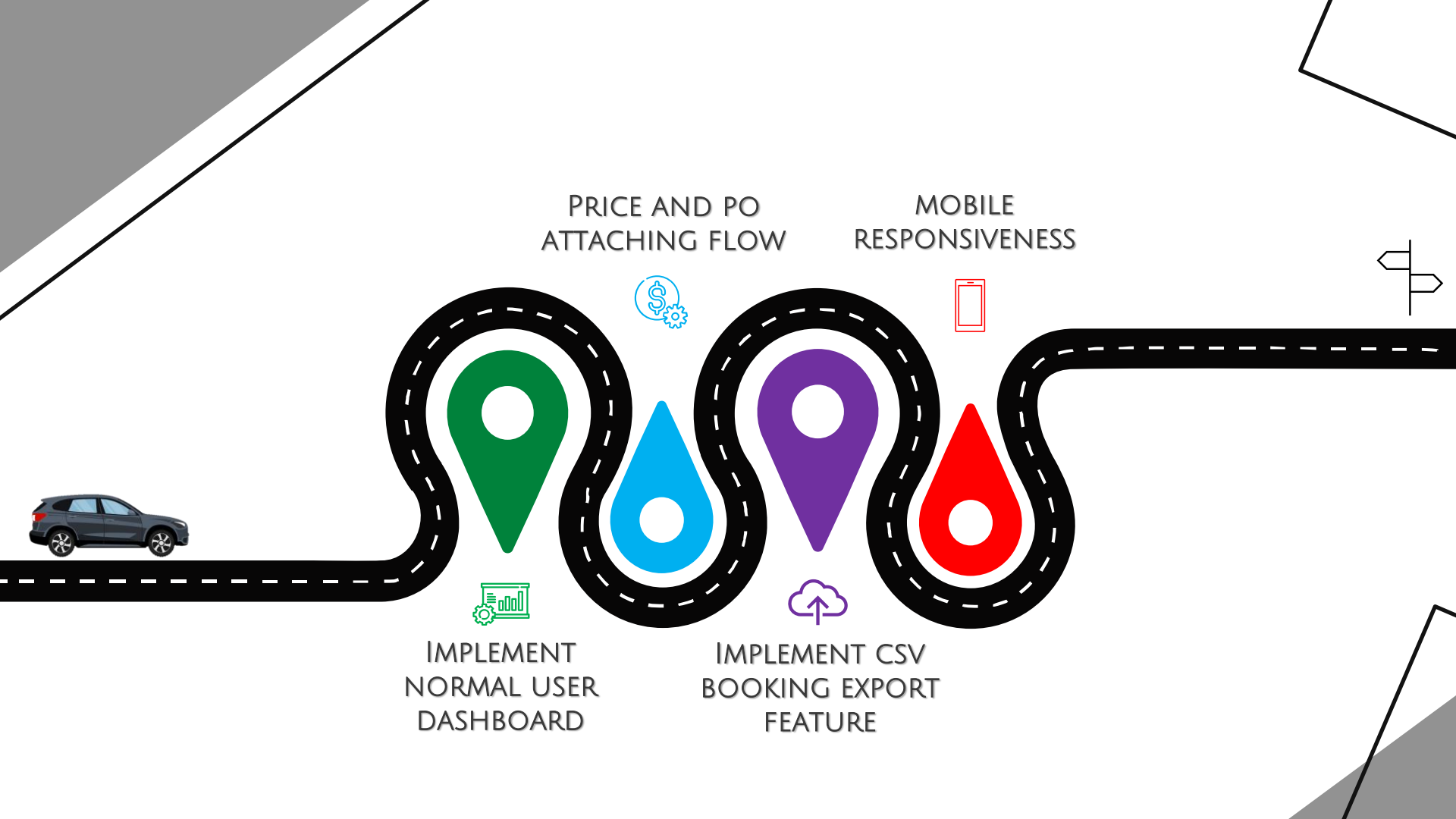
Every user completed five tasks, and each task was categorised into four groups: Log-in/Log-out, Booking Creation, Booking Approval/Rejection, and Miscellaneous tasks.

The Booking Creation tasks varied between users; some were asked to complete airport transfers, while others performed less complicated bookings. Similarly, the Booking approval/rejection tasks varied between users, with some users being given different sets of information used to locate a booking. An additional miscellaneous task was added to test functions of the system outside of the primary functionality; this included tasks such as asking the user to change their profile's phone number. The task list can be found in the project repository [here](#).

When taking part in the study, the users were prompted to use the think-aloud protocol, narrating their thoughts while their voices were captured for future analysis. This provided qualitative data and highlighted the points at which the application didn't align with the user's expectations, causing friction.

From the screen captures of each participant's session, quantitative data was extracted in the form of task completion rates and the time taken per task. Task completion was divided into three categories: success, success with assistance, and failure. This categorisation will aid in identifying workflows within the application that need further consideration. Additionally, the time taken per task will allow for comparisons between task categories, facilitating further optimization of user flows for the tasks taking the longest time.

When conducting the study, strong ethical standards were followed. This included providing all participants with an information sheet outlining the study's requirements. Once read and understood, participants were given time to ask questions or express any concerns they might have regarding their participation in the study. Participants were then asked to sign a consent form confirming their willingness to participate and informing them of their right to withdraw from the study at any time. All participant's data is anonymised with no record of participants' personal details being stored alongside the data collected. Participants may however be recognised by voice; this risk was clearly outlined in the documentation which all participants agreed to. These procedures ensured that the study followed university ethical frameworks and data protection standards.



PRICE AND PO
ATTACHING FLOW

MOBILE
RESPONSIVENESS

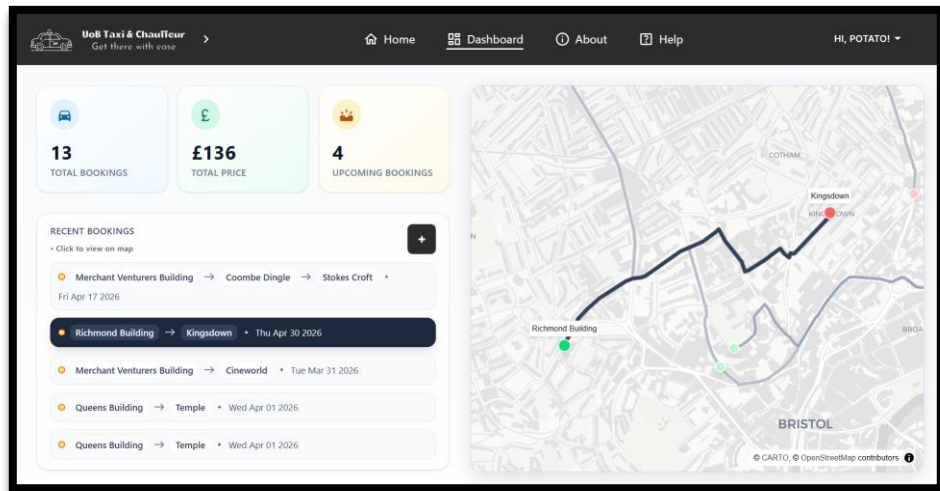
IMPLEMENT
NORMAL USER
DASHBOARD

IMPLEMENT CSV
BOOKING EXPORT
FEATURE



IMPLEMENT NORMAL USER DASHBOARD

Booking Status	Price	Operation		
Pending	Add	View	Edit	Cancel
Approved	£ 46	View		
Pending	Add	View	Edit	Cancel
Pending	£ 67	View	Edit	Cancel
Rejected	£ 12	View		
Pending	Add	View	Edit	Cancel



Attach a price

Enter a price below to be attached to the booking:

Price
56

CLOSE ATTACH

Attach PO number

Please enter the PO number below to approve the booking.

PO number

CLOSE ATTACH



PRICE AND PO ATTACHING FLOW





IMPLEMENT CSV BOOKING EXPORT FEATURE



Economics... Selected B...

Management Panel

Export Bookings From

Export by department
Choose a department to export the bookings.

Ready to export
Export all 56 bookings [Clear](#)

☒	Name	Pick-up Time
☒	Soggy Waffles	4/4/2026, 10:40:00 AM
☒	Potato Chip	4/17/2026, 11:30:00 AM
☒	Potato Chip	4/30/2026, 12:30:00 PM
☒	Fyodor Tolstol	4/18/2026, 9:22:00 PM
☒	Potato Chip	3/31/2026, 8:22:00 PM

Select a department to export

- Chemistry 69
- Biochemistry 9
- Economics 5
- Computer Science 4
- Policy Studies 3

Export by department
Choose a department to export the bookings.

Policy Studies

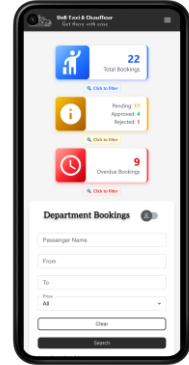
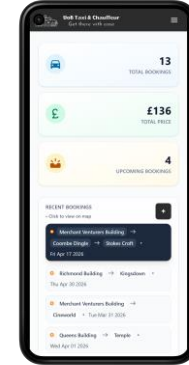
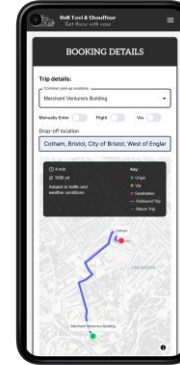
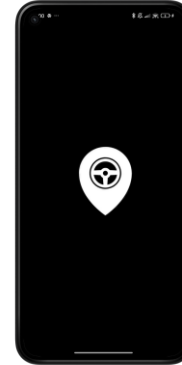
[Export all Policy Studies bookings](#)



PWA
(INSTALLABLE ON
MOBILE & DESKTOP)

Booking Approved
Your booking has been approved. Click to view details.

Price Attachment
A price of £97 has been attached to a booking by the admin. Click to view details.



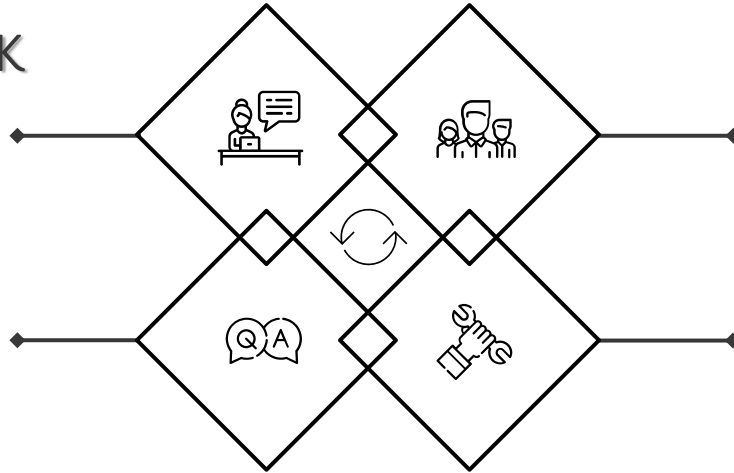
MOBILE
RESPONSIVENESS
AND
COMPATIBILITY



OUR PROCESS

MENTOR FEEDBACK

Identify weak points, areas of improvement to iterate on



TEAM MEETING

Follow agenda, track progress, assign tasks,

PUBLISH WORK

Create a PR for review, have discussion on certain features, Merge

WORK ON ISSUE

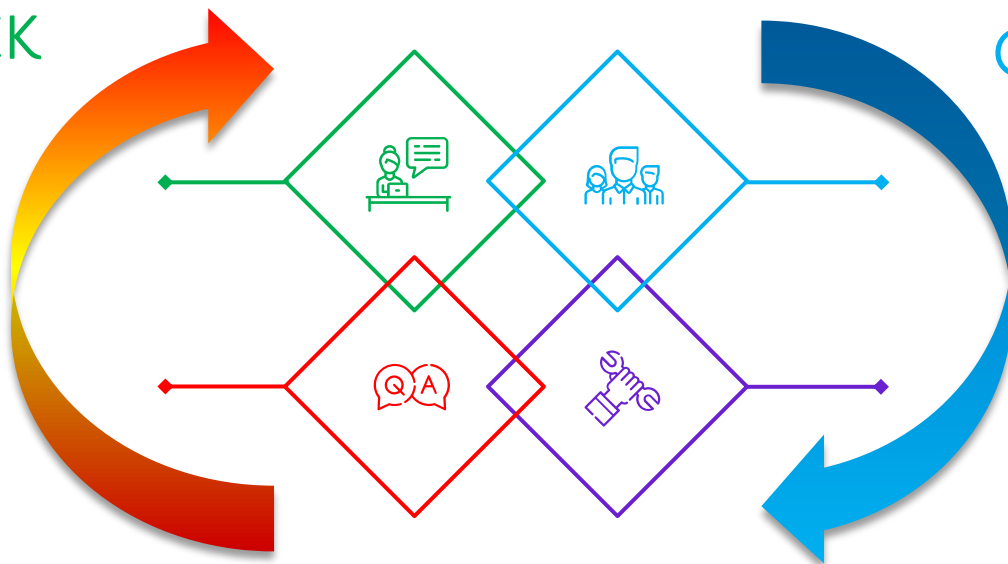
Work on the task, develop, ask help if needed



AGILE DEVELOPMENT

FEEDBACK

COLLABORATE



ITERATE

DEVELOP



ERIK MALTBY

- Smooth client communication
- His **connection** with university staff and students led to more **testing opportunities**
- Great stand-up **leader** that made meetings **effective** and **fun**.

VALUES BEYOND METRICS



ALAN YANG

- Brought in new **perspective** and scolded us a bit...
- Reason for the **creation** of the meeting **agenda**
- Our reliable **test writer** and TDD **lead**

CHALLENGES & SOLUTIONS

01

Unstructured team meetings
resulting in drawn out, ineffective
meeting



vs



Appoint a rotating stand-up
leader, using pre-defined
agenda

01

CHALLENGES & SOLUTIONS

01

Unstructured team meetings
resulting in drawn out, ineffective
meeting

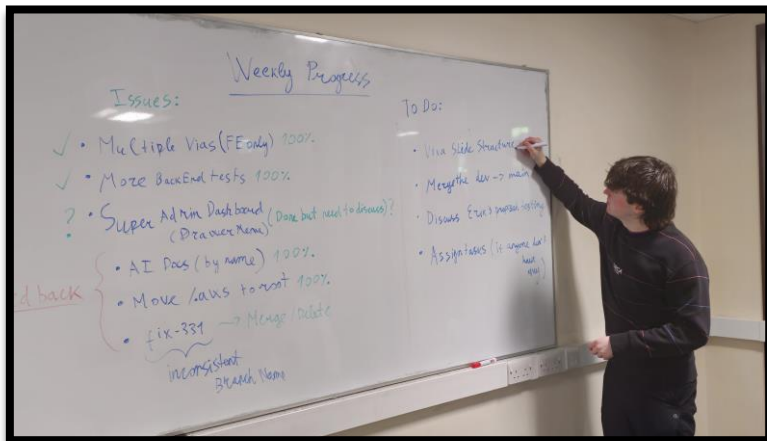


vs



Appoint a rotating stand-up
leader, using pre-defined
agenda

01



Overview

1. Recap of goals from previous sprint

Team Update

1. What was accomplished since last meeting?
2. What are the current tasks?
3. What's planned for the upcoming week?

Blocks

1. Any challenges affecting progress?
2. Who can help fix those?

Review

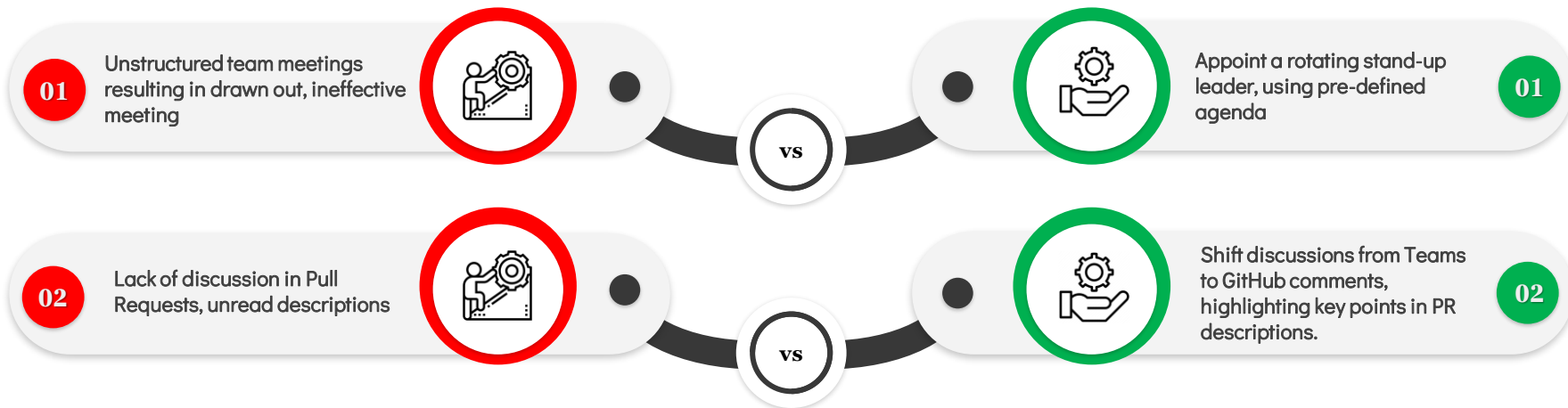
1. Progress on backlog items
2. Adjust priorities for next sprint

Actions

1. Summarize takeaways
2. Assign responsibilities

Any other business

CHALLENGES & SOLUTIONS



CHALLENGES & SOLUTIONS

 **ioanmoir** commented 3 weeks ago • edited

Related issue numbers:
[#304](#)

 Important

You will need to make some changes to your local repo if this PR is merged:

- `npx prisma db pull` (or switching connection keys to the updated DB branch).
- Add the `OSRM_SECRET` token to your `.env` file.

 **Otgon-Erdene-Otgonsukh** commented 3 days ago • edited

Related issue numbers:
[#249](#) [#377](#)

Description

 Important

This PR contains essential information about what accounts to use with different roles in development since after protecting the pages, we are no longer can just use 1 account to access every page. I will send the credentials of each account with their emails to the teams chat when the PR is ready.

Refactor DB - Delete useless columns - Add constraints ✓ **Back-End** **Database** 1 19

#317 by Dehors777 was merged on Feb 11 • Approved 5 of 14 tasks Beta Release

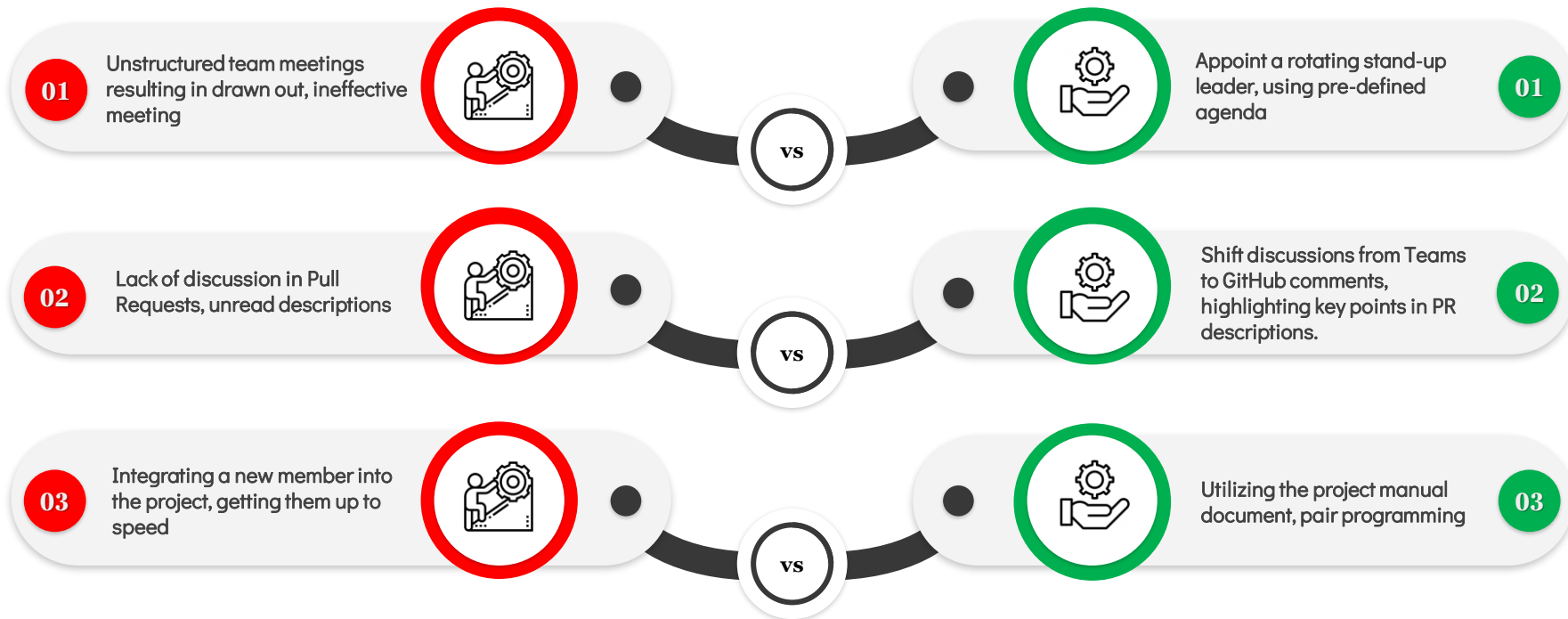
Department Manage Section in Super Panel ✓ **enhancement** **UI/UX** 1 17

#395 by Dehors777 was merged 2 weeks ago • Approved 8 of 14 tasks Final Release

Profile page implementation with Back End logic ✓ **Back-End** **Database** **UI/UX** 2 15

#254 by Otgon-Erdene-Otgonsukh was merged on Jan 26 • Approved 9 of 14 tasks Beta Release

CHALLENGES & SOLUTIONS



CHALLENGES & SOLUTIONS

returning all names for bookings ...



emaltby and **Otgon-Erdene-Otgonsukh** committed last week

added a assign manager api call in frontend ...



emaltby and **Otgon-Erdene-Otgonsukh** committed yesterday

added a unassign manager to frontend ...



emaltby and **Otgon-Erdene-Otgonsukh** committed yesterday

added api route for assigning and unassigning managers ...



emaltby and **Otgon-Erdene-Otgonsukh** committed yesterday

Project manual & guide

Description

This manual provides information on the tools used in the development of this project, including brief descriptions that explain the reasons for choosing these tools, as well as instructions on how to utilize them. The goal of this guide is to enable team members who are facing difficulties understanding certain architectural concepts of the tools involved to get up to speed or for new developers who are not familiar with the tools involved with concise and clear documentation that addresses all these issues.

- [Next.js](#)
- [Neon](#)
- [Prisma](#)
- [Jest](#)
- [CI/CD](#)
- [AWS](#)

Next.js (Javascript Framework)





The website **design**
and **user experience**
is **consistent** and
simple



The final product
ticked all the
requirement boxes
and **beyond**



The **detailed**
documentation will
make future
development clear

POSITIVES

CLIENT FEEDBACK

FUTURE PLANS



Planned to keep in
touch for small future
developments



Market the product with the
University to **adopt** it as the
main system



Arrange an informal
dinner meeting to
chat about our
experiences etc.

05

SEP RECAP

What just happened ?

WHAT WE LEARNED

TECHNICAL SKILLS

Learning new frameworks, languages and actually applying them to a real-life project.

HAVING A RESPONSIBILITY

Being part of a team gave us a sense of responsibility, set expectations, and pushed us to be reliable.

REAL-LIFE APPLICABLE SKILLS

Communicating with a real client about the project improved our communication skills and gave us industrial insight.

RESILIENCE

Working on a long-running project taught us the importance of staying resilient and consistent despite challenges



MORE PAIR PROGRAMMING

More in-person and collaborative coding sessions would have ensured everyone was up to speed



STRUCTURED MEETINGS

Follow a defined agenda meetings early on to get the fullest out of team meetings



DATABASE STRUCTURE

Clearly defining the database structure early would have prevented major code changes later.



EQUAL WORKLOAD

Focusing on supporting teammates to maintain a balanced workload, even if it slows progress on new tasks.

WHAT WE WOULD DO DIFFERENTLY



SEP ROUND 2

THANK
YOU !



ANY
QUESTIONS ?